

# PeerLink Database Schema

## Account & Access

Django user plus security, notification, and audit records

| auth_user   |                     |
|-------------|---------------------|
| * id        | SMALLINT PK         |
| username    | VARCHAR(150) UNIQUE |
| email       | VARCHAR(254)        |
| first_name  | VARCHAR(150)        |
| last_name   | VARCHAR(150)        |
| password    | VARCHAR(128)        |
| is_staff    | BOOLEAN             |
| is_active   | BOOLEAN             |
| date_joined | TIMESTAMP           |

| user_security_state  |             |
|----------------------|-------------|
| * id                 | SMALLINT PK |
| user_id              | FK UNIQUE   |
| must_change_password | BOOLEAN     |

| notification |              |
|--------------|--------------|
| * id         | SMALLINT PK  |
| user_id      | FK           |
| message      | VARCHAR(255) |
| is_read      | BOOLEAN      |
| action_tab   | VARCHAR(50)  |
| created_at   | TIMESTAMP    |

| audit_log  |             |
|------------|-------------|
| * id       | SMALLINT PK |
| user_id    | FK NULL     |
| action     | VARCHAR(32) |
| model_name | VARCHAR(64) |
| object_id  | VARCHAR(64) |
| created_at | TIMESTAMP   |

**Notes**  
auth\_user is the single identity table.  
Deleting a user cascades to security state,  
notifications, profiles, posts and comments.  
audit\_log keeps rows after user deletion (SET NULL).

1:1

1:N

1:N

# PeerLink Database Schema

## Profile Core

Mentor and mentee profiles with their competency bridge tables

### interest\_tag

|      |             |        |
|------|-------------|--------|
| * id | SMALLINT    | PK     |
| name | VARCHAR(50) | UNIQUE |

### mentor\_profile

|              |              |        |
|--------------|--------------|--------|
| * id         | SMALLINT     | PK     |
| user_id      | FK           | UNIQUE |
| program      | VARCHAR(100) |        |
| year_level   | SMALLINT     |        |
| gpa          | DECIMAL(3,2) | NULL   |
| bio          | TEXT         |        |
| skills       | JSON         |        |
| availability | JSON         |        |
| capacity     | SMALLINT     |        |
| gender       | VARCHAR(10)  |        |
| approved     | BOOLEAN      |        |

### mentee\_profile

|                  |              |        |
|------------------|--------------|--------|
| * id             | SMALLINT     | PK     |
| user_id          | FK           | UNIQUE |
| program          | VARCHAR(100) |        |
| year_level       | SMALLINT     |        |
| campus           | VARCHAR(100) |        |
| student_id_no    | VARCHAR(10)  |        |
| contact_no       | VARCHAR(11)  |        |
| learning_style   | VARCHAR(100) |        |
| difficulty_level | SMALLINT     |        |
| preferred_gender | VARCHAR(20)  |        |
| approved         | BOOLEAN      |        |

### Notes

Both profiles are 1:1 with auth\_user.  
interest\_tag is shared through two M2M  
join tables created by Django.  
competency\_id points at the Curriculum page.

### mentor\_competency

|                            |          |     |
|----------------------------|----------|-----|
| * id                       | SMALLINT | PK  |
| mentor_id                  | FK       |     |
| competency_id              | FK       |     |
| proficiency_level          | SMALLINT | 1-5 |
| UNIQUE(mentor, competency) |          |     |

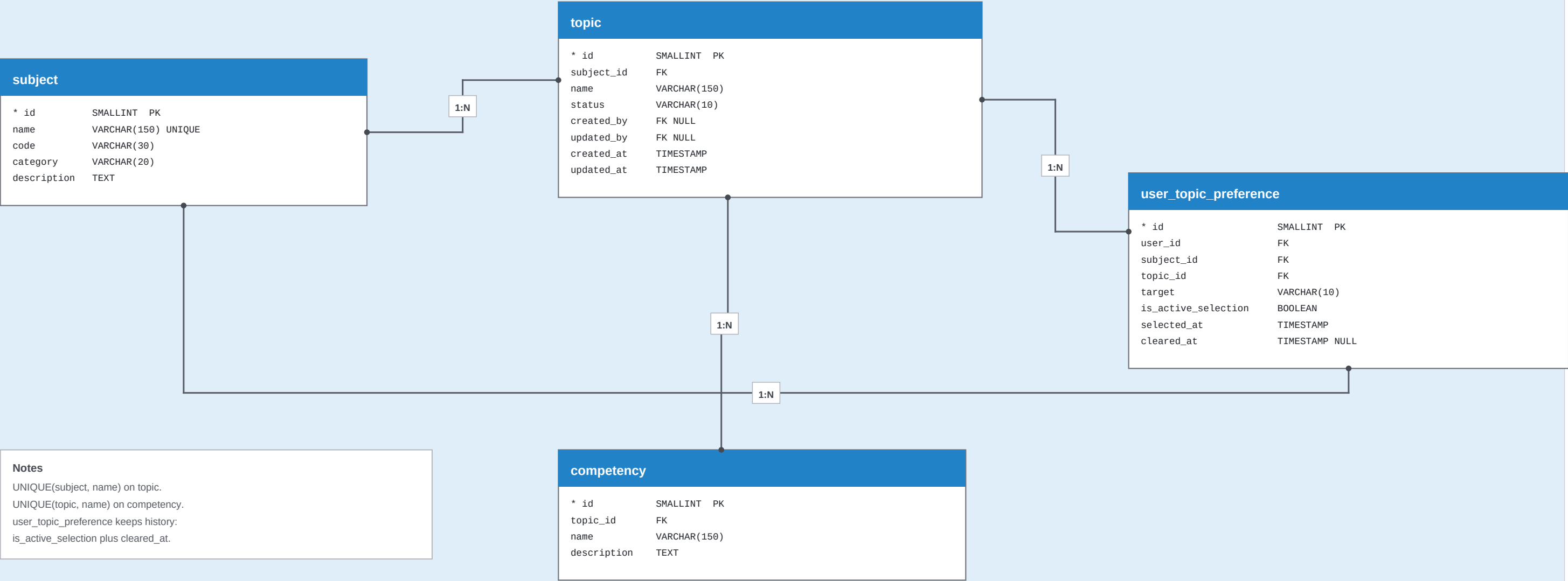
### mentee\_competency\_need

|                            |          |     |
|----------------------------|----------|-----|
| * id                       | SMALLINT | PK  |
| mentee_id                  | FK       |     |
| competency_id              | FK       |     |
| need_level                 | SMALLINT | 1-5 |
| UNIQUE(mentee, competency) |          |     |

# PeerLink Database Schema

## Curriculum Graph

Subject to topic to competency, plus per-user topic selections



# PeerLink Database Schema

## Matching & ML

Pairing requests, training dataset, and model registry

| mentee_mentor_request  |           |      |
|------------------------|-----------|------|
| * id                   | SMALLINT  | PK   |
| mentee_id              | FK        |      |
| mentor_id              | FK        |      |
| created_at             | TIMESTAMP |      |
| accepted               | BOOLEAN   |      |
| accepted_at            | TIMESTAMP | NULL |
| UNIQUE(mentee, mentor) |           |      |

| matching_dataset_record |             |      |
|-------------------------|-------------|------|
| * id                    | SMALLINT    | PK   |
| mentee_id               | FK          | NULL |
| mentor_id               | FK          | NULL |
| label                   | BOOLEAN     |      |
| source                  | VARCHAR(16) |      |
| features                | JSON        |      |
| created_at              | TIMESTAMP   |      |

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| matching_dataset_topic |              |      |
|------------------------|--------------|------|
| * id                   | SMALLINT     | PK   |
| dataset_record_id      | FK           |      |
| topic_id               | FK           | NULL |
| subject_id             | FK           | NULL |
| role                   | VARCHAR(10)  |      |
| topic_snapshot         | VARCHAR(150) |      |

### Notes

External keys, see other pages:

mentee\_id, mentor\_id -> Profile Core

topic\_id, subject\_id -> Curriculum Graph

created\_by -> auth\_user

Dataset rows survive profile deletion

because both FKs are SET NULL.

### model\_metadata

|                             |              |        |
|-----------------------------|--------------|--------|
| * id                        | SMALLINT     | PK     |
| version                     | VARCHAR(64)  | UNIQUE |
| status                      | VARCHAR(16)  |        |
| artifact_path               | VARCHAR(255) |        |
| metrics                     | JSON         |        |
| feature_names               | JSON         |        |
| training_rows               | INTEGER      |        |
| topic_count_active_at_train | INTEGER      |        |
| created_by                  | FK           | NULL   |

# PeerLink Database Schema

## Community & Feed

Mentor announcements, comments, and the social post feed

| announcement |           |      |  |
|--------------|-----------|------|--|
| * id         | SMALLINT  | PK   |  |
| mentor_id    | FK        |      |  |
| message      | TEXT      |      |  |
| created_at   | TIMESTAMP |      |  |
| deleted_at   | TIMESTAMP | NULL |  |

| announcement_recipient     |          |    |  |
|----------------------------|----------|----|--|
| * id                       | SMALLINT | PK |  |
| announcement_id            | FK       |    |  |
| user_id                    | FK       |    |  |
| UNIQUE(announcement, user) |          |    |  |

| user_post  |             |           |  |
|------------|-------------|-----------|--|
| * id       | SMALLINT    | PK        |  |
| author_id  | FK          |           |  |
| text       | TEXT        |           |  |
| image      | IMAGE       | NULL      |  |
| category   | VARCHAR(20) |           |  |
| likes      | M2M         | auth_user |  |
| created_at | TIMESTAMP   |           |  |

| post_comment |           |    |  |
|--------------|-----------|----|--|
| * id         | SMALLINT  | PK |  |
| post_id      | FK        |    |  |
| author_id    | FK        |    |  |
| content      | TEXT      |    |  |
| created_at   | TIMESTAMP |    |  |

| comment         |           |    |  |
|-----------------|-----------|----|--|
| * id            | SMALLINT  | PK |  |
| author_id       | FK        |    |  |
| announcement_id | FK        |    |  |
| content         | TEXT      |    |  |
| created_at      | TIMESTAMP |    |  |

### Notes

announcement uses soft delete (deleted\_at).

No recipient rows means every accepted mentee of that mentor can see the post.

mentor\_id -> Profile Core page.

author\_id and user\_id -> auth\_user.

# PeerLink Database Schema

## Cross-Module Map

How auth\_user connects the five areas of the database

